Lecture 1 — Disorders of Haemostasis & Thrombosis

Companion Figures

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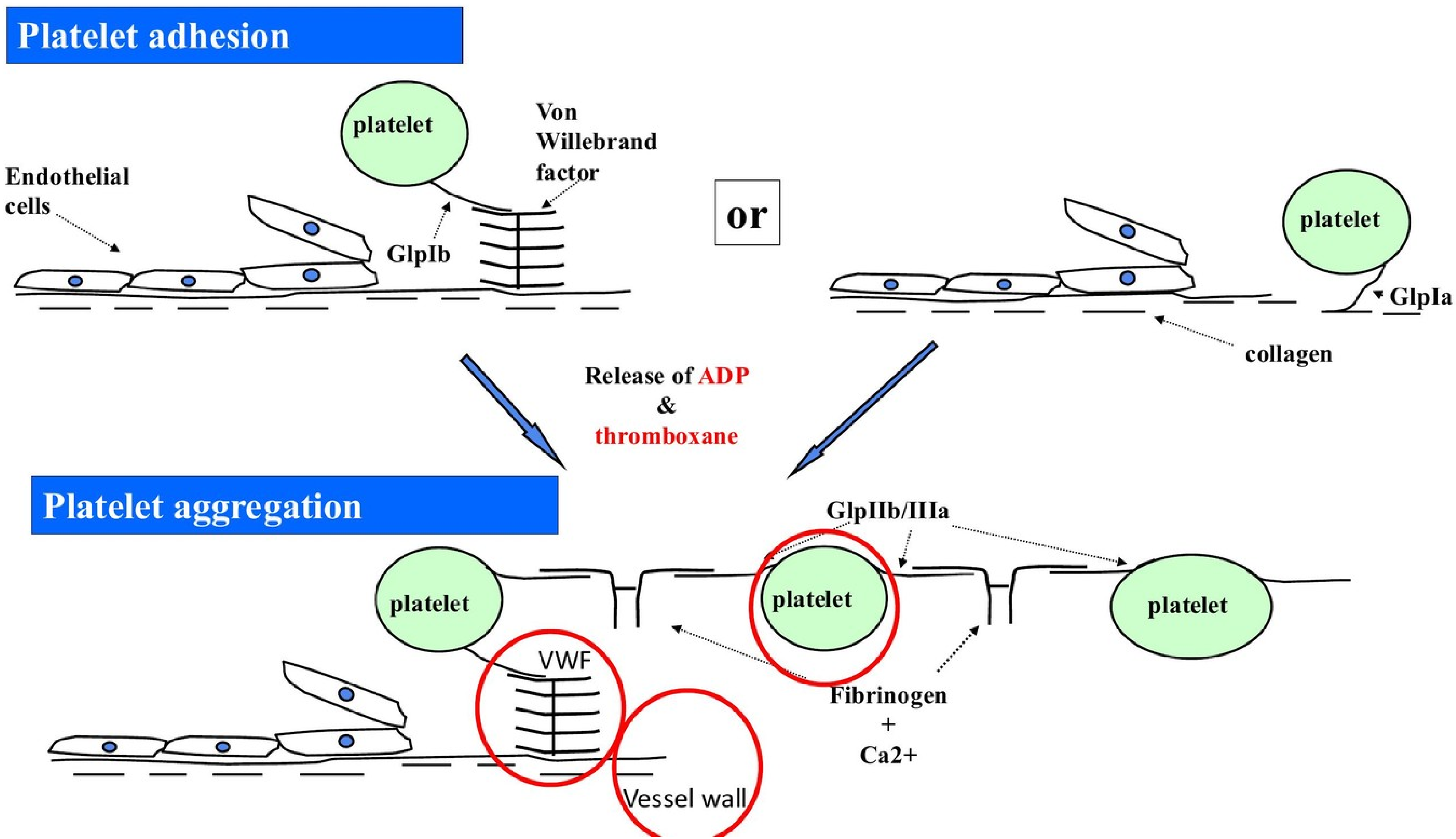
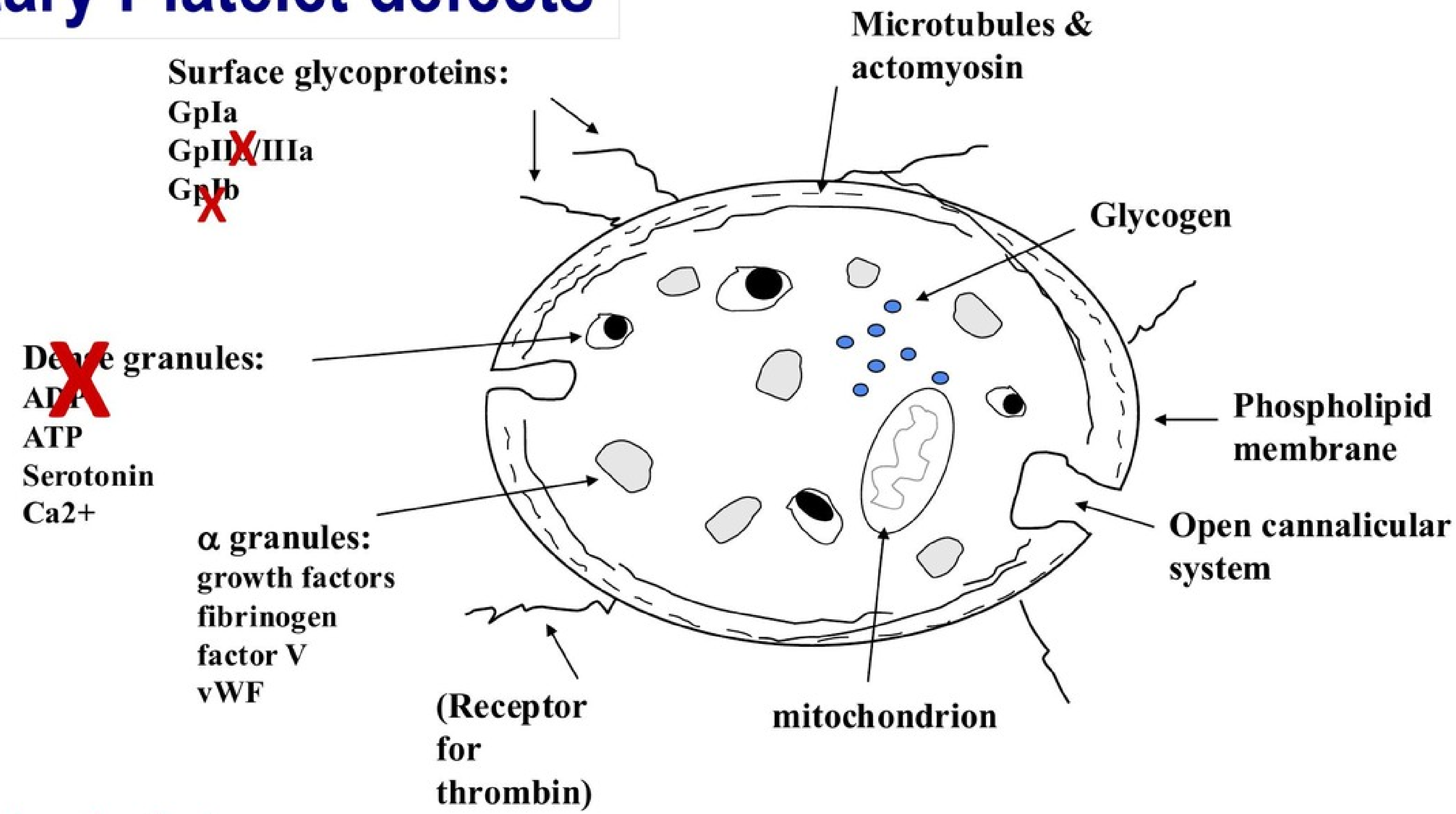


Figure L1.1 — Platelet adhesion and aggregation pathways

Hereditary Platelet defects



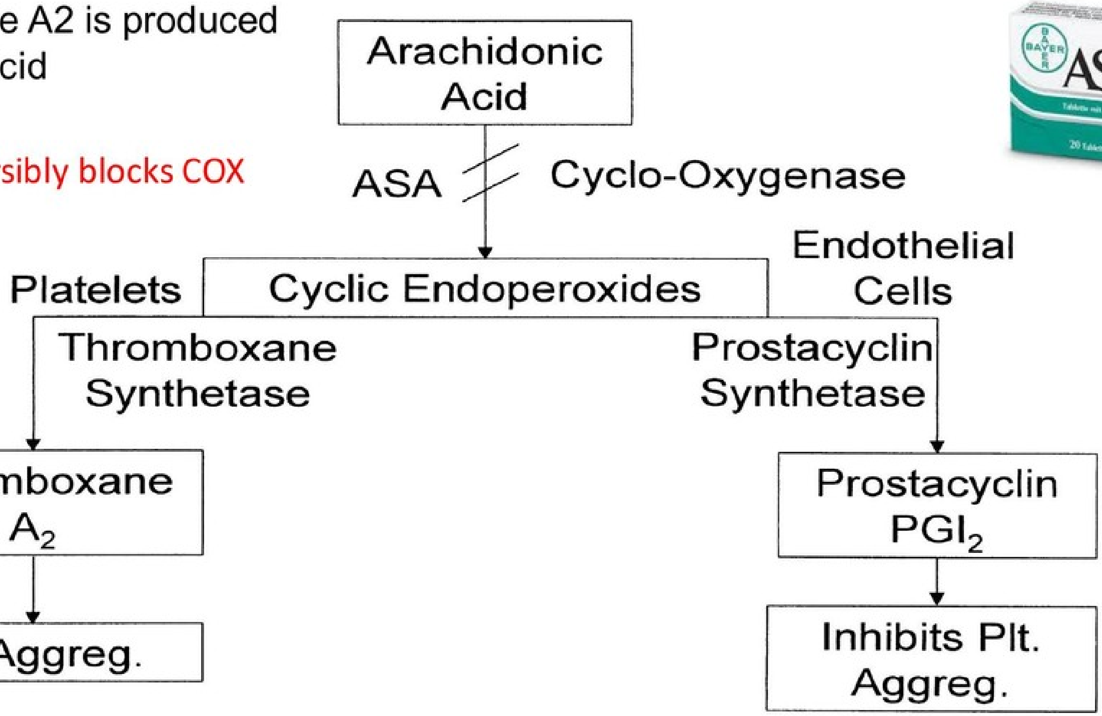
Glanzmann's thrombasthenia
Bernard Soulier syndrome
Storage Pool disease

Figure L1.2 — Platelet anatomy and hereditary platelet defects (Glanzmann GpIIb/IIIa, Bernard-Soulier GpIb, storage pool dense granules)

Antiplatelet therapy: widely used in the prevention and treatment of cardiovascular & cerebrovascular disease

The prostaglandin Thromboxane A2 is produced by platelets from Arachidonic acid

Aspirin irreversibly blocks COX



Clopidogrel irreversibly blocks the ADP receptor on platelets

1-BRS-CVR-10 Haematological disorders: Summarise the pathology and pathophysiology of haematological disorders.

1-BRS-CVR-11 Haematological disorders: Describe the clinical features and treatment options of haematological disorders

Figure L1.3 — Arachidonic acid pathway: aspirin's mechanism on COX; platelet thromboxane vs endothelial prostacyclin

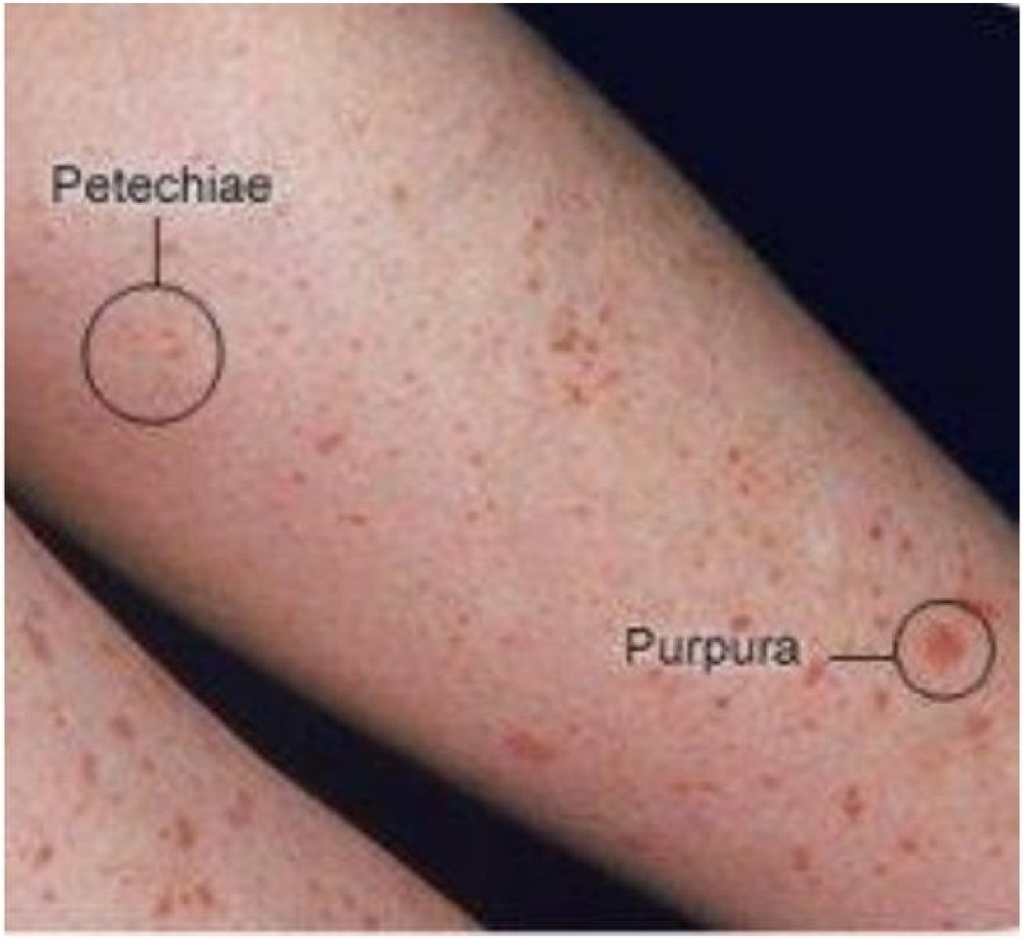


Figure L1.4 — Petechiae and purpura on skin

Clinical features of disorders of primary haemostasis

Petechiae and
Purpura in a
Patient with Vasculitis

Ecchymosis and
purpura in
a patient with
immune
thrombocytopenia

'Senile' purpura

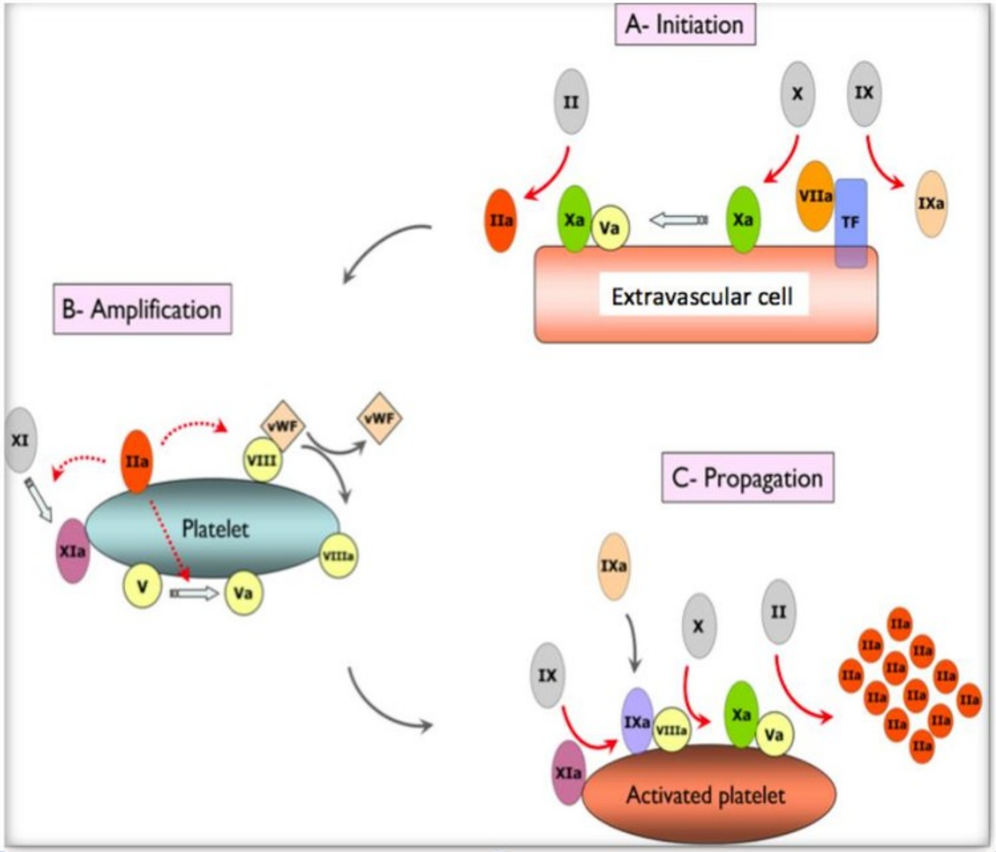
Easy bruising (ecchymosis)

Petechiae and Purpura in a
Patient with Acute Myeloid Leukaemia

Increased skin
elasticity in connective tissue disorders

Figure L1.5 — Clinical appearances: petechiae/purpura in vasculitis, AML, ITP; senile purpura; ecchymosis; hyperelastic skin

and hence fibrin formation



s: Summarise the pathology and pathophysiology of haematological disorders

s: Describe the clinical features and treatment options of haematological disorders

Figure L1.6 — Cell-based coagulation model (initiation, amplification, propagation)

Figure L1.7 — Acute haemarthrosis (swollen knee, hallmark of haemophilia)

Figure L1.8 — Chronic haemarthrosis with muscle wasting in untreated haemophilia

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Figure L1.9 — Haematoma from IM injection in haemophilia patient

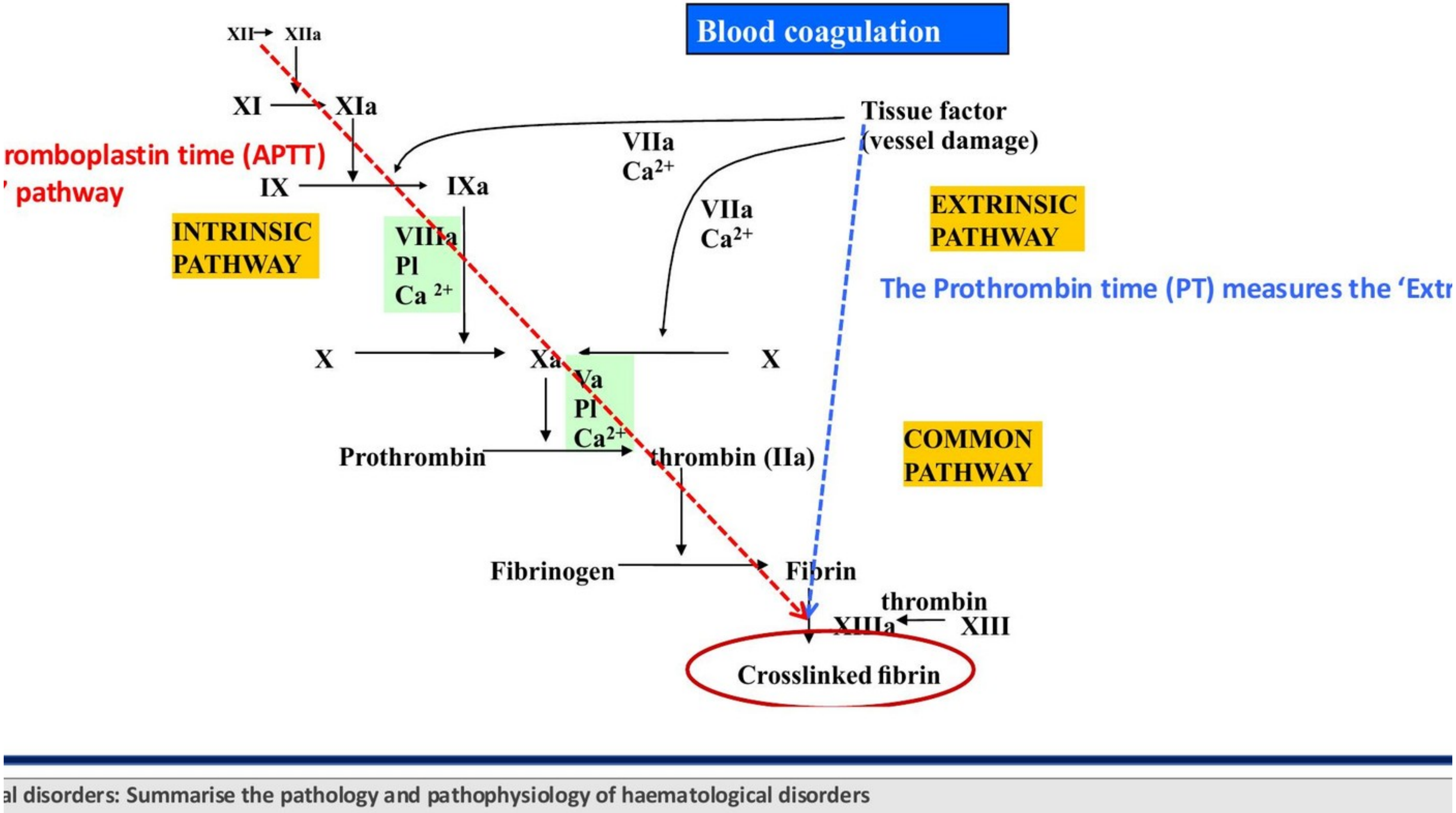
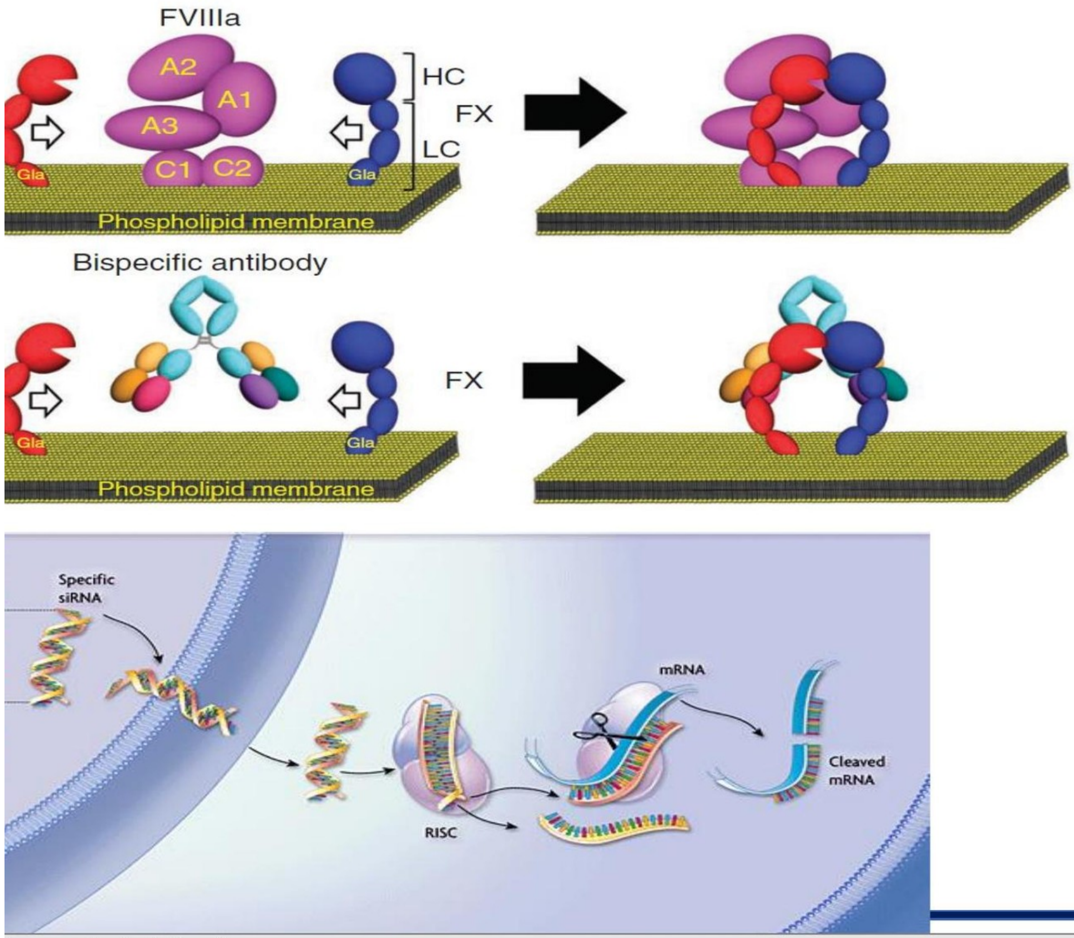


Figure L1.10 — Coagulation cascade showing APTT (intrinsic) vs PT (extrinsic) pathways

aemophilia



haematological disorders

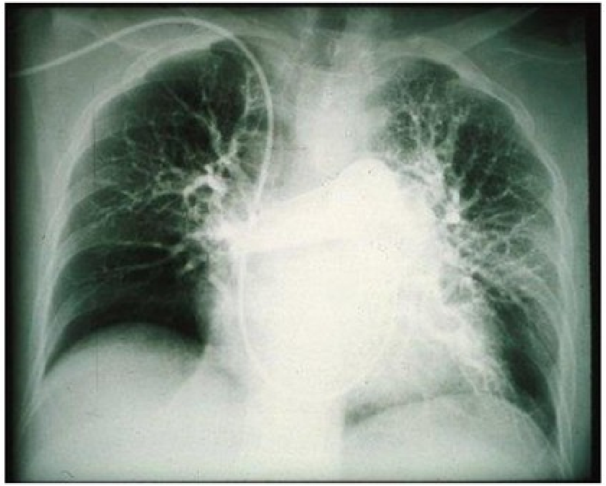
ıs of haematological disorders

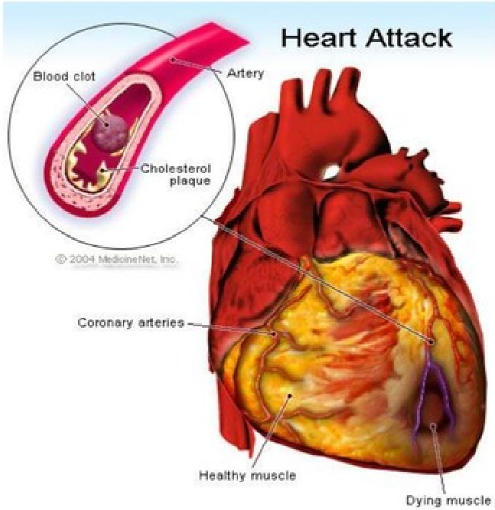
Figure L1.11 — Novel haemophilia therapies: emicizumab bispecific antibody bridging FIXa and FX; RNA silencing of antithrombin

How do Disorders of Thrombosis present?

Pulmonary Embolism (PE)

- Tachycardia
- Hypoxia
- Shortness of breath
- Chest pain
- Haemopysis
- Sudden death





Most people die with a 'haemostatic end-point'

Deep Vein Thrombosis (DVT)

- Painful leg
- Swelling
- Red
- Warm
- May embolise to lungs
- Post thrombotic syndrome

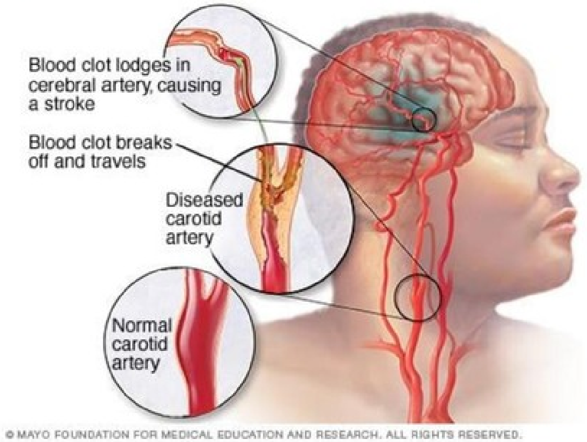


Figure L1.12 — Presentations of thrombosis: PE (CXR), DVT (swollen red leg), MI, stroke

nticoagulant proteins

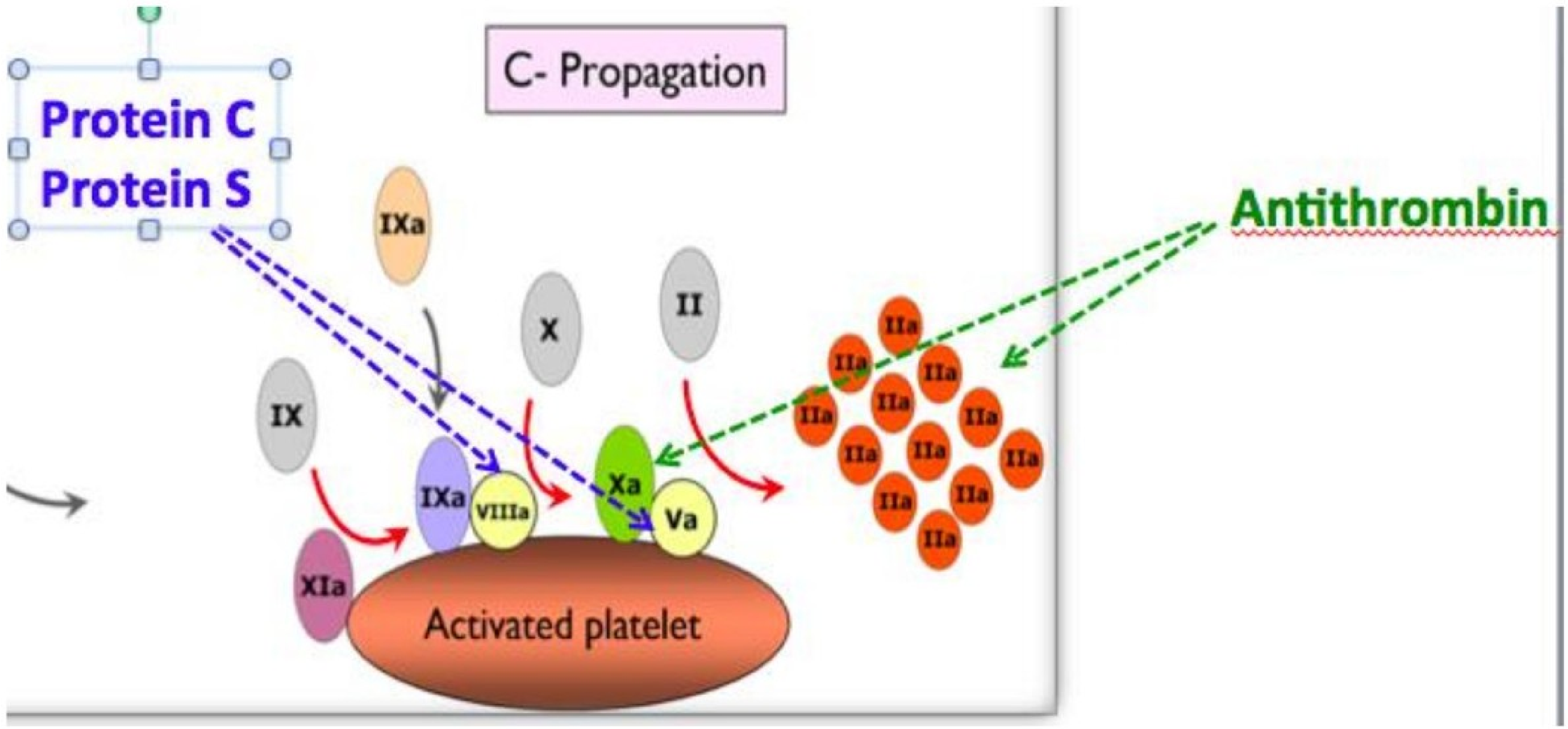


Figure L1.13 — Natural anticoagulant proteins: Protein C + S inactivate Va and VIIIa; antithrombin inactivates thrombin and Xa

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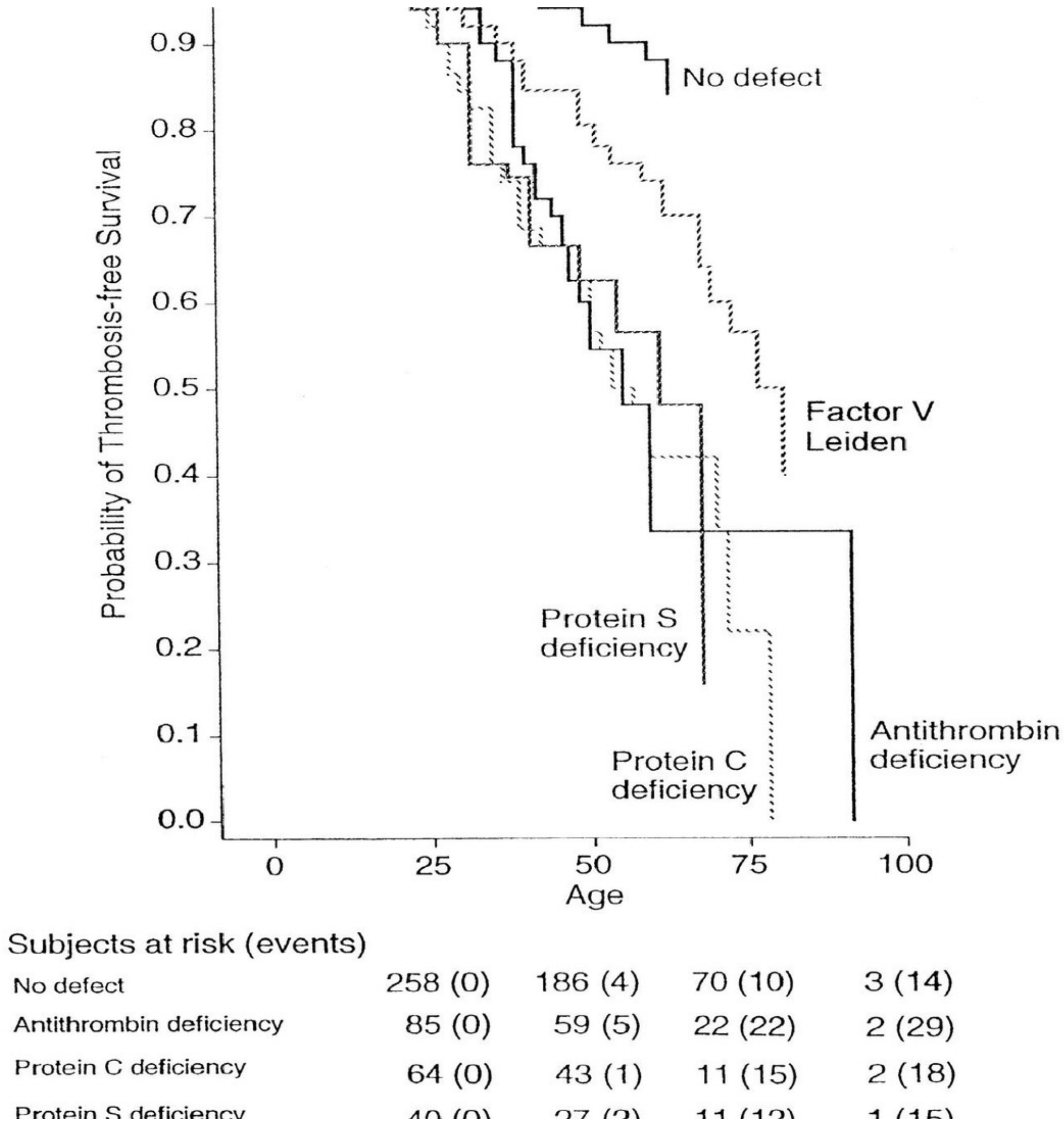
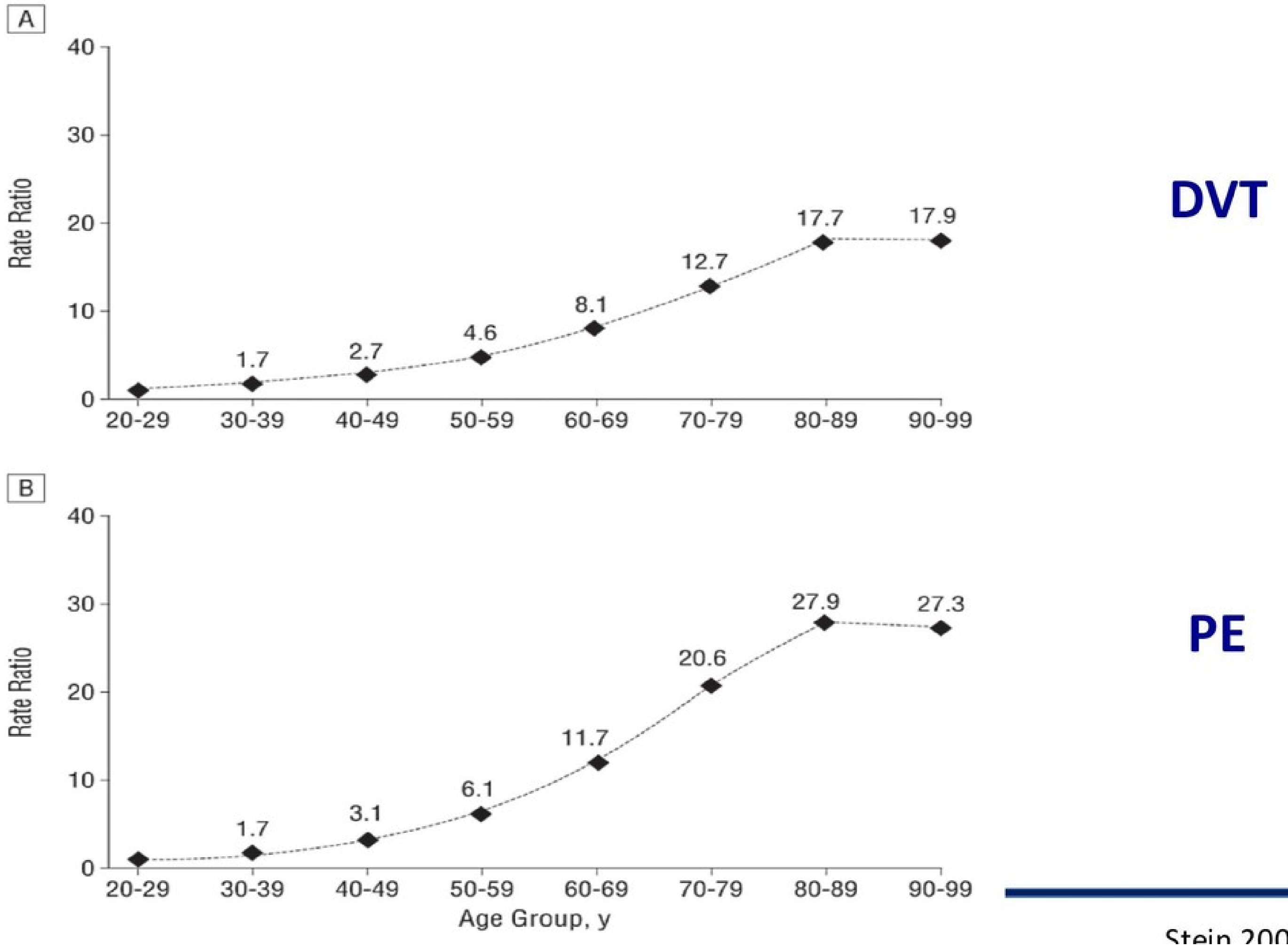


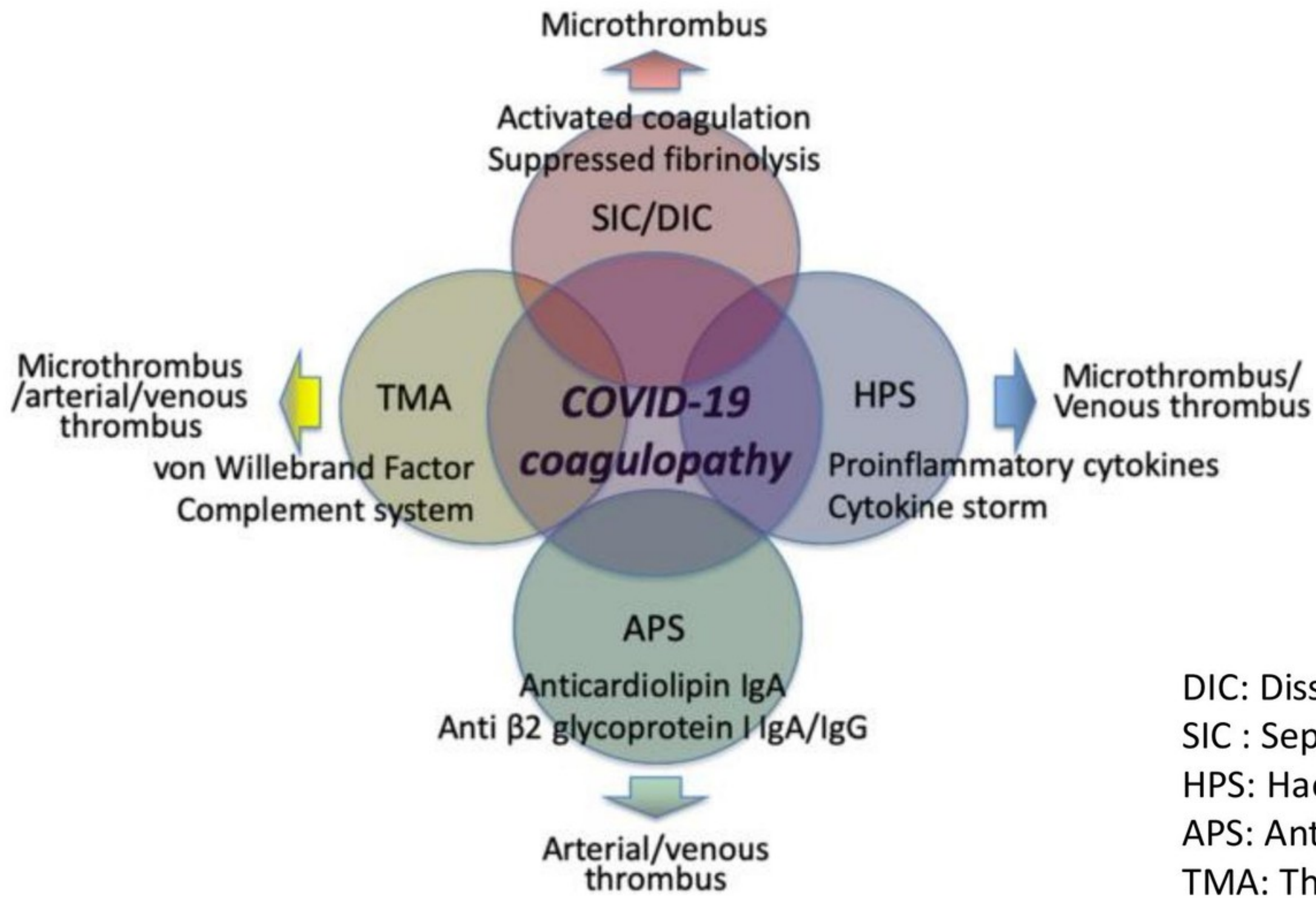
Figure L1.14 — Thrombosis-free survival by inherited thrombophilia (Martinelli 1998): AT deficiency has steepest drop

Venous thrombosis and age



Stein 2004

Figure L1.15 — Rate ratio of DVT and PE by age (Stein 2004): rises ~17× (DVT) and ~28× (PE) by 80s



DIC: Disseminat
SIC : Sepsis-Indu
HPS: Haemopha
APS: Antiphospl
TMA: Thrombot

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Figure L1.16 — SARS-CoV-2 coagulopathy: overlapping SIC/DIC, TMA, HPS, APS mechanisms

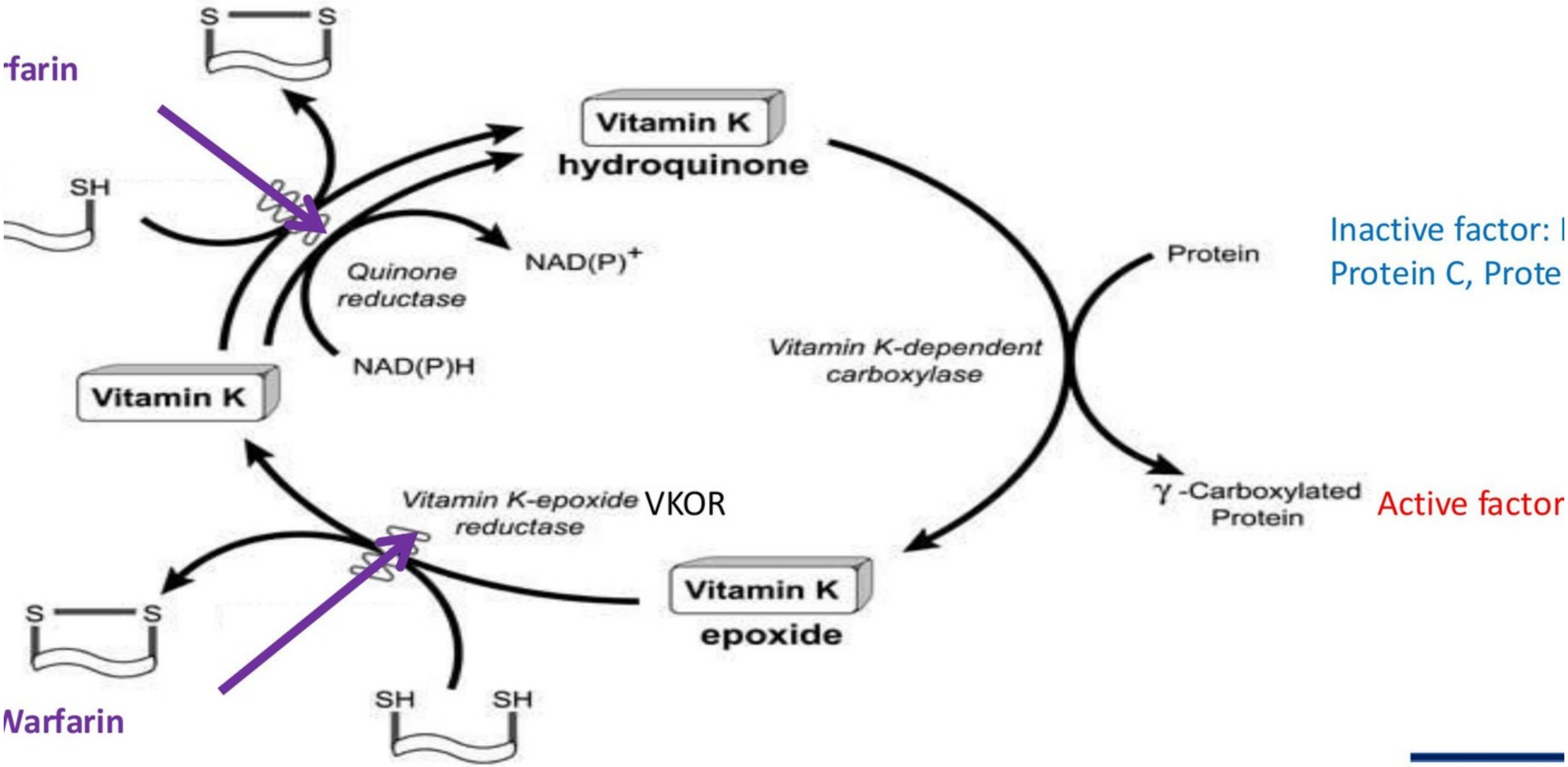

Figure L1.17 — Vitamin K cycle: warfarin inhibits VKOR, preventing regeneration of reduced vitamin K needed for γ-carboxylation of II, VII, IX, X, Protein C and S

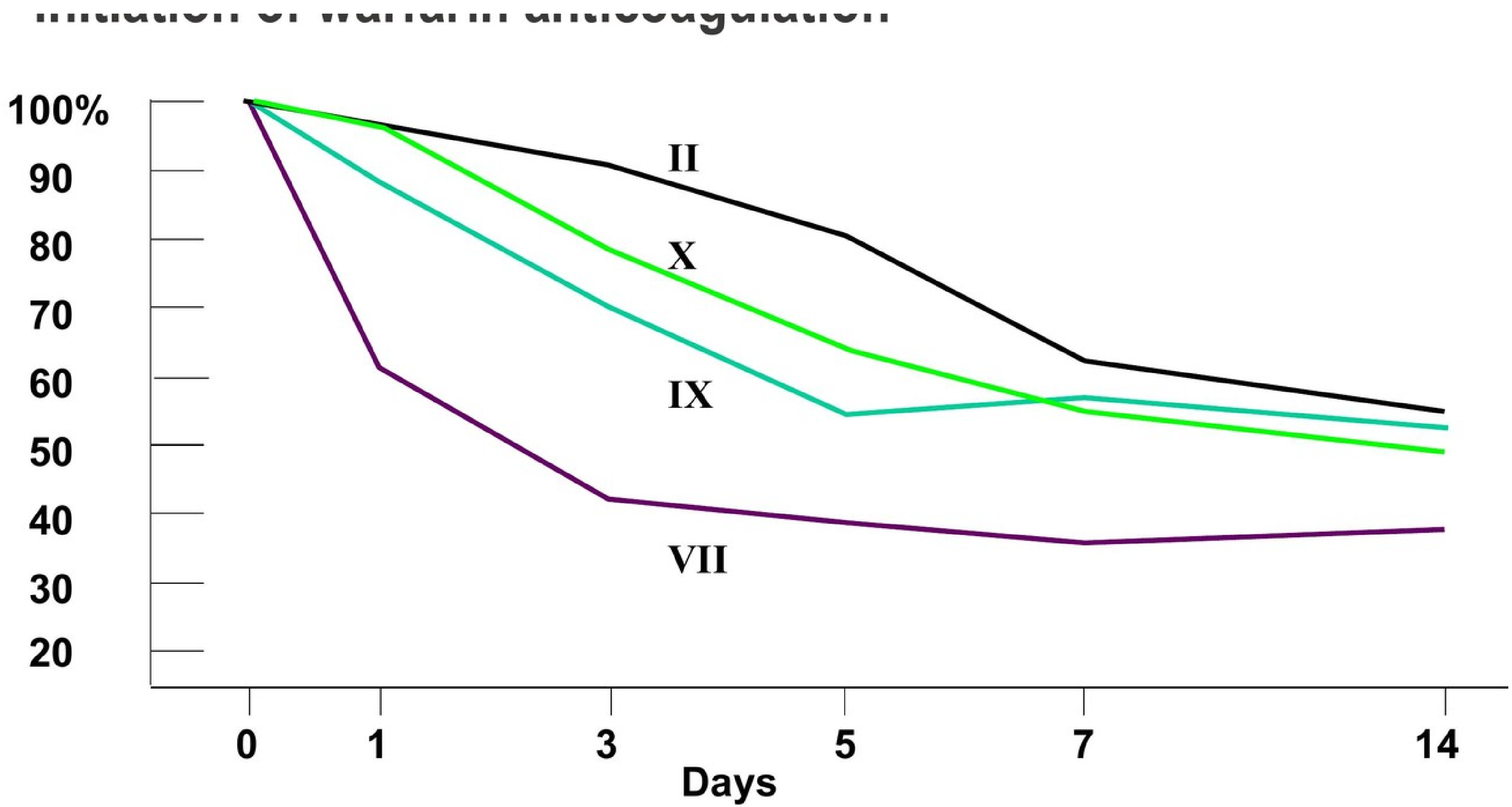

Figure L1.18 — Time course of factor depletion on warfarin: VII (shortest half-life) falls fastest; II (longest) falls slowest — explains skin-necrosis risk if not heparin-bridged

Skin Necrosis
Severe Protein C deficiency
2-3 days after starting Warfarin
Thrombosis predominantly in adipose tissues

ata

Purple toe syndrome
Disrupted atheromatous plaques bleed
Cholesterol emboli lodge in extremities

ders
disorders

Figure L1.19 — Warfarin side effects: skin necrosis (Protein C deficiency; adipose-tissue thrombosis) and purple toe syndrome (cholesterol emboli)

Chondrodysplasia Punctata

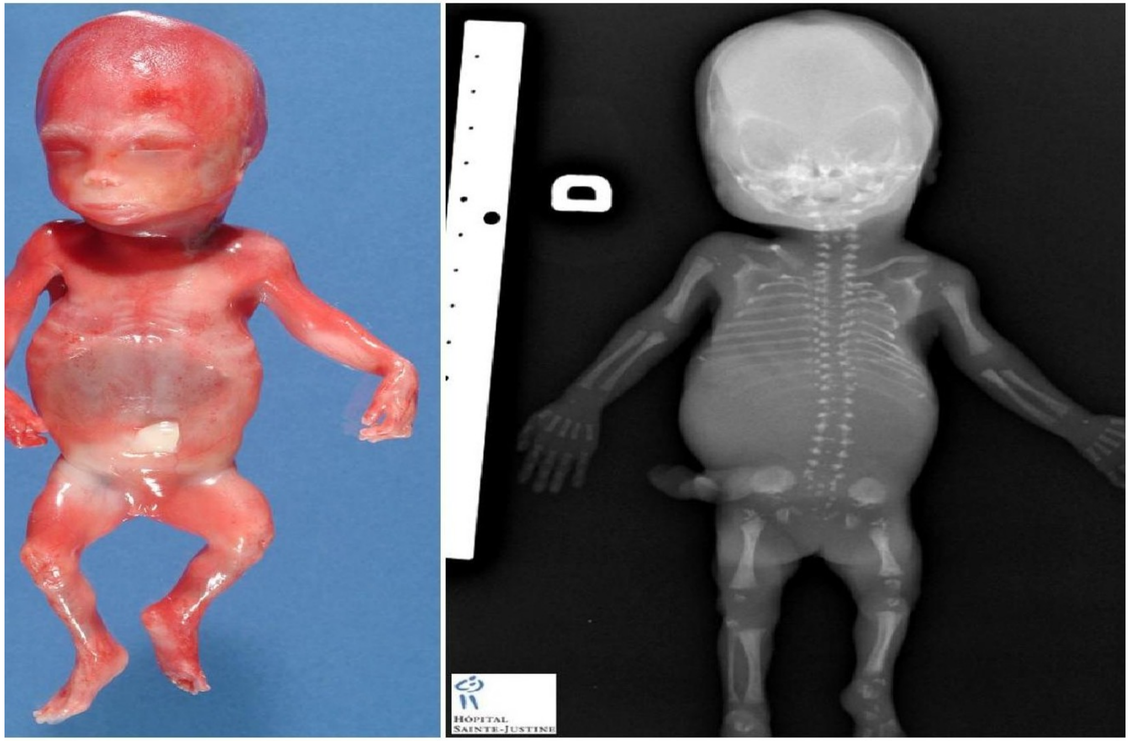

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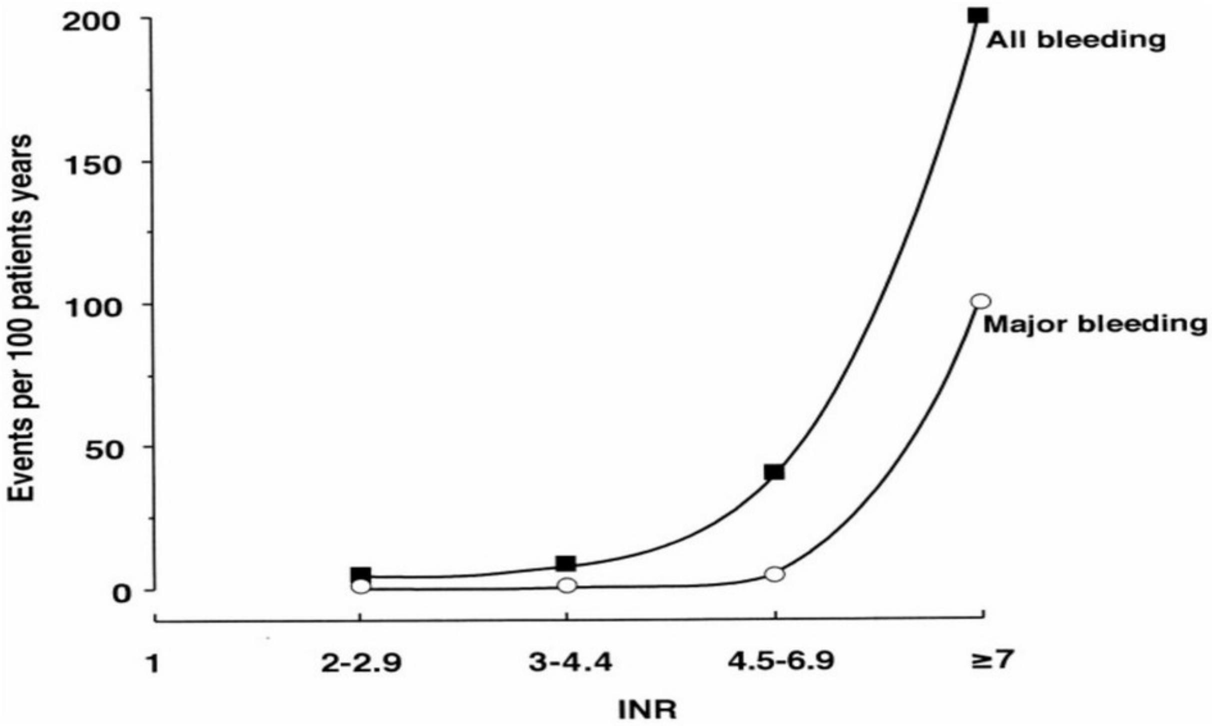
Figure L1.20 — Chondrodysplasia punctata: warfarin embryopathy; early epiphyseal fusion

›leeding risk vs. INR



Makris 200

Figure L1.21 — Bleeding events per 100 patient-years vs INR (Makris 2001): steep rise above INR 4

- Enhancement of Antithrombin
    - Inactivation of thrombin (Hep binds AT + Thrombin)
    - Inactivation of FXa (Hep binds AT only)
    - (Inactivation of FIXa, FXIa, FXIIa)

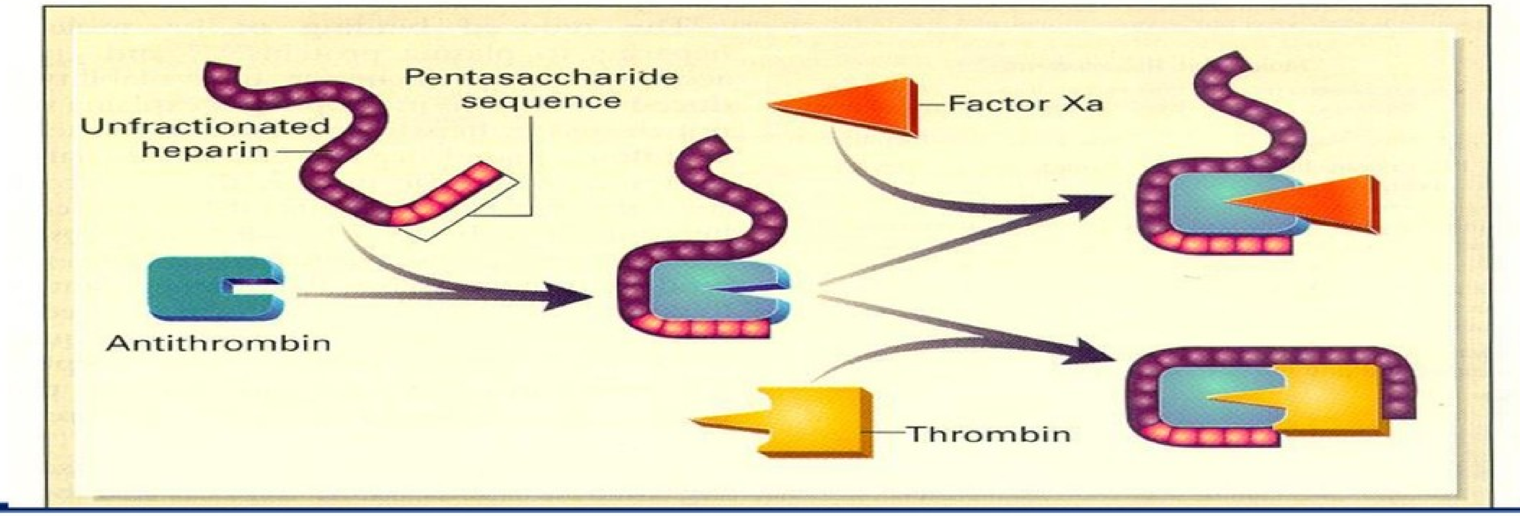
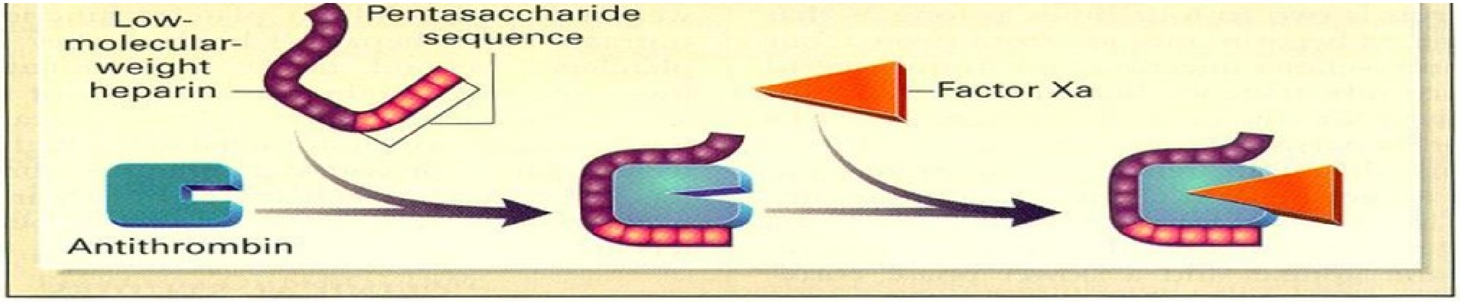


Figure L1.22 — Unfractionated heparin: pentasaccharide binds antithrombin, activating it against Xa; long chain additionally wraps thrombin



- Contain pentasaccharide sequence for binding AT
- Predictable dose response in most cases so does not require monitoring (cf UFH)

Figure L1.23 — Low-molecular-weight heparin: pentasaccharide binds antithrombin → mainly inactivates Xa; chain too short to bridge thrombin

Direct Oral Anticoagulants (DOACs)

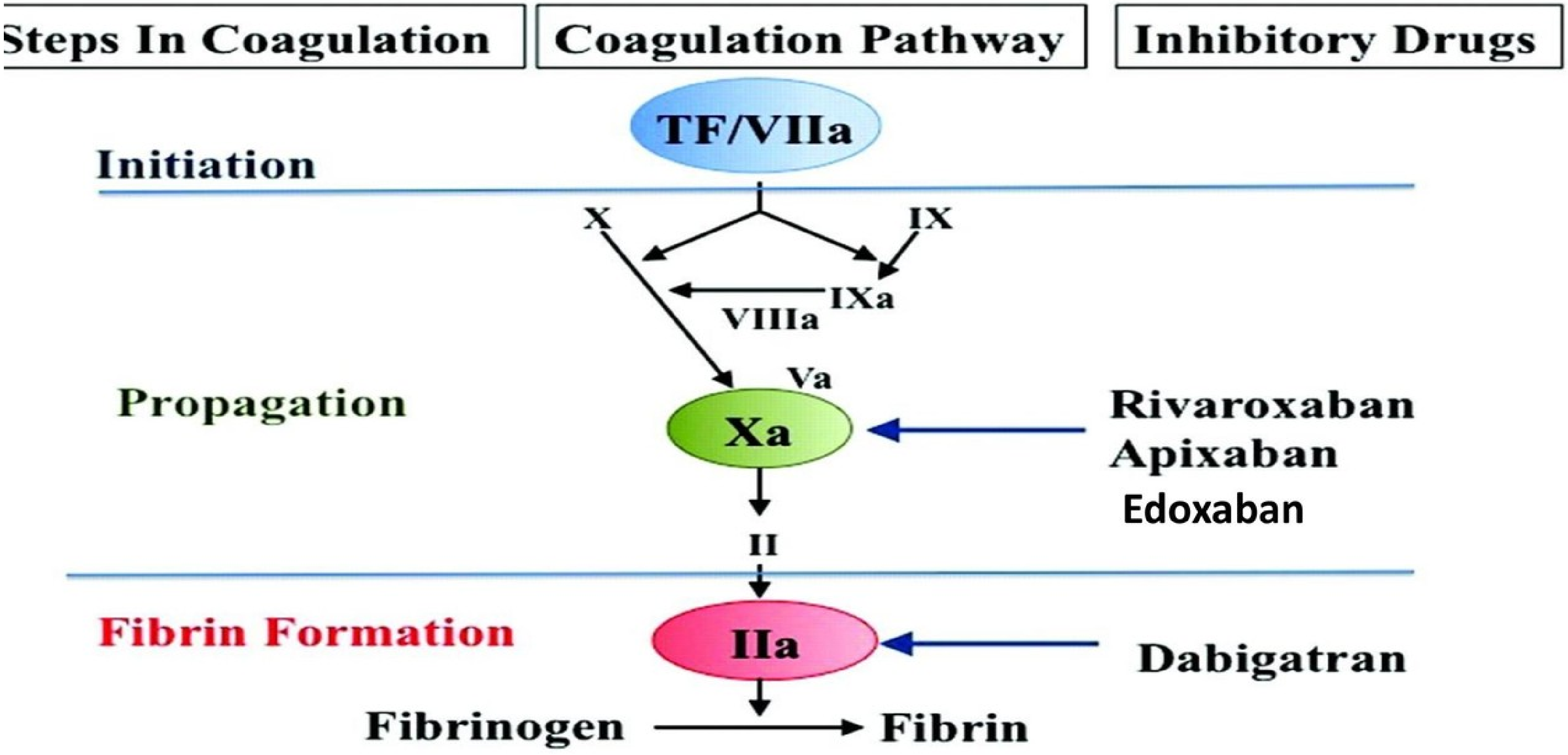


Figure L1.24 — DOAC targets: rivaroxaban, apixaban, edoxaban inhibit Xa; dabigatran inhibits IIa (thrombin)